

palsr: Projected Actor Locations for spatial modeling of dyadic interactions between moving actors

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Software

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Summary

Many actors studied in the social sciences have no fixed geographic location. Armed groups roam across a theater of conflict, firms relocate, diplomats and international organizations shift their operational focus, and migrating populations move in response to events. For such *moving actors*, the location at which two of them interact is itself a meaningful, modelable outcome, but standard spatial-interaction tools assume actors sit at fixed points and so cannot be applied directly. The Projected Actor Location (PALS) method (Kim et al. 2023) estimates where a mobile actor effectively “is” at any point in time from the spatiotemporal history of its past interactions, then uses these projected locations to model both the distance between pairs of actors and the probability that they interact.

`palsr` is an R package (R Core Team 2024) that implements the PALS workflow: constructing and validating dyadic event data, estimating the smoothing parameters by minimizing great-circle prediction error, projecting actor locations at arbitrary times, building the dyadic distance covariates used in downstream interaction models, and quantifying uncertainty through a nonparametric bootstrap with multiple-imputation (Rubin’s Rules) pooling. The performance-critical kernels (the Haversine distance and the projection smoother) are implemented in C++ through `Rcpp` (Eddelbuettel 2013), so that parameter estimation, which repeatedly re-projects every actor across a grid of times, runs quickly even on data sets with thousands of events.

Statement of need

The empirical study of geopolitical interaction (alliances, conflict, cooperation, trade) is overwhelmingly *dyadic*: the unit of analysis is a pair of actors, and a central predictor of whether and how two actors interact is the distance between them (Gleditsch and Ward 2001; Ward et al. 2007). When actors are states with fixed capitals or centroids, this distance is trivial to compute. But many substantively important actors are mobile: rebel groups, militias, terrorist organizations, peacekeeping deployments, and non-governmental organizations. The fine-grained, geocoded event data that now document them, such as the Armed Conflict Location and Event Data project (Raleigh et al. 2010), record *where each interaction happened* but assign actors no persistent coordinates. Treating these actors as fixed-location entities discards exactly the spatial dynamics that drive their interactions.

Existing R spatial software does not do this. General geospatial toolkits such as `sf` (Pebesma 2018) and spatial point-process libraries such as `spatstat` (Baddeley and Turner 2005) operate on objects with given coordinates; they provide no way to *infer* a moving actor’s effective location, let alone one tuned to predict future interactions. Conversely,

dyadic interaction models in political science and economics typically take inter-actor distance as an exogenous input. Prior to PALS there was thus no general, estimable method for assigning time-varying locations to mobile actors in a way that is optimized for predicting their interactions.

PALS was built for this setting, and the original article (Kim et al. 2023) shows that PALS-based distances substantially improve the prediction of subnational conflict relative to naive location measures. That contribution, however, came only with replication scripts specific to a single application. `palsr` packages the method as documented, tested software that others can reuse. Applied researchers in political science, conflict studies, economics, and geography can apply PALS to their own dyadic event data without re-deriving the estimator; the package propagates estimation uncertainty into downstream models through an event bootstrap, and it ships both the original ACLED Nigeria data and a simulated-data generator, so the full workflow can be learned and reproduced out of the box. To our knowledge it is the first general-purpose software implementation of projected-location modeling for moving actors.

Method

For a focal actor i at prediction time t , PALS forms a recency-weighted mean of the locations of i 's own past events (the *focal* component) and a recency-weighted mean of the locations of events involving i 's past interaction partners, or *alters* (the *alter* component). The projected location is a convex combination of the two,

$$g_i(t) = (1 - \pi) \sum_e W_i(e) g(e) + \pi \sum_e W_k(e) g(e), \quad \pi = \text{logistic}(\gamma + \eta v),$$

where event weights decay with age as in exponential smoothing (Gardner Jr. 2006), governed by α for the focal actor and β for the alters, and the mixing weight π depends through γ and η on the focal actor's activity relative to its alters, summarized by the event-count statistic v . The four parameters are estimated by “marching forward” through time: every event is predicted using only events strictly preceding it, and parameters are chosen to minimize the mean Haversine distance between predicted and observed interaction locations. The package also provides a one-parameter variant that fixes $\pi = 0$ and estimates only α , which is fast and frequently competitive.

Because the projected location is recomputed as time advances, each actor traces a *trajectory* through space rather than occupying a fixed point. Figure 1 illustrates this for four actors in the bundled simulated data: PALS projects each actor's location at yearly intervals, and the resulting paths drift through the cloud of observed events.

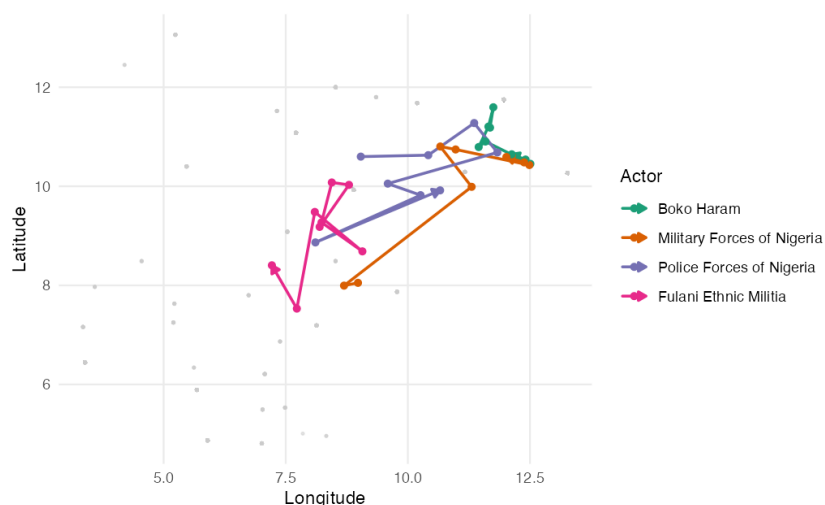


Figure 1: PALS-projected locations of four major actors in the `nigeria_acled` ACLED data (Boko Haram, the Nigerian military and police, and the Fulani Ethnic Militia), fit with the one-parameter model and projected at yearly intervals from 2009 to 2016. Grey points are the observed conflict events; coloured paths trace each actor's projected trajectory (arrows point forward in time).

Key features

- **Validated data container** (`pal_events()`) for dyadic, time-stamped, georeferenced events.
- **Parameter estimation** (`estimate_pals()`) for both the full four-parameter and the reduced one-parameter models, via Haversine-error minimization with a compiled objective.
- **Projection** (`project_pal()`, `project_pals()`) of actor locations at any set of times.
- **Prediction and covariate construction** (`predict_event_locations()`, `pal_distance()`) for interaction locations and dyadic distances, including a log transform for conflict-style specifications.
- **Uncertainty quantification** (`bootstrap_pals()`, `pool_rubin()`) via a nonparametric event bootstrap and Rubin's Rules (Rubin 1987).
- **Reproducible example data** (`nigeria_acled`, `simulate_conflict_events()`), a vignette, and a full test suite that also checks the C++ kernels against pure-R reference implementations.

Example

```
library(palsr)
data(nigeria_acled) # 1,549 ACLED conflict events (Nigeria)

fit <- estimate_pals(nigeria_acled, model = "one") # estimate alpha
coef(fit)

# Project where each actor is on a given date.
project_pals(nigeria_acled,
             predict_time = as.Date("2015-12-01"),
             params = fit)

# Build the dyadic distance covariate for an interaction model.
```

```
dyads <- data.frame(
  actor1 = "Boko Haram",
  actor2 = "Fulani Ethnic Militia (Nigeria)",
  time   = as.Date("2014-06-01")
)
pal_distance(nigeria_acled, dyads, fit, transform = "log")

# Bootstrap uncertainty in the smoothing parameter.
summary(bootstrap_pals(nigeria_acled, R = 10, model = "one", seed = 1))
```

Availability and quality control

`palsr` is developed openly on GitHub. The package ships with documentation for every exported function, an introductory vignette that walks through the full workflow on `nigeria_acled`, and a `testthat` suite covering data validation, estimation, projection, distance construction, and uncertainty quantification. The tests additionally verify the compiled C++ kernels against independent pure-R reference implementations, and continuous integration runs `R CMD check` across Linux, macOS, and Windows on every change.

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